Spoken Tutorial – Understanding AI, Machine Learning, and Deep Learning

Title: Understanding AI, Machine Learning, and Deep Learning

Module	Introduction to Artificial Intelligence
Episode	AI, ML, and Deep Learning: The Core Concepts
Learning Objective	At the end of this tutorial learner will be able to 1. Understand the core definition and goal of Artificial Intelligence. 2. Differentiate between Artificial Intelligence, Machine Learning, and Deep Learning. 3. Explain the fundamental difference between traditional programming and Machine Learning. 4. Identify real-world applications of Deep Learning.
Approx. Duration	10 minutes
Outline	• Introduction to Artificial Intelligence (AI) • Defining Machine Learning (ML) • ML vs. Traditional Programming: A Cat Analogy • Understanding Deep Learning (DL) • Applications of Deep Learning
Meta Tags	AI, Artificial Intelligence, Machine Learning, ML, Deep Learning, DL, Neural Networks, Spoken Tutorial, Computer Science
Pre-requisite Tutorial	Basic computer literacy and an understanding of what software programs do.

Script

Visual Cue	Narration
	Welcome to this **Spoken Tutorial** on **Artificial Intelligence**, **Machine Learning**, and **Deep Learning**. Here, we will understand these important topics and their differences.
	After this tutorial, you will learn several key points. You will understand what **Artificial Intelligence** means. You will differentiate between **AI**, **Machine Learning**, and **Deep Learning**. You will learn how **Machine Learning** differs from traditional programming. You will also identify real-world applications of **Deep Learning**.
	To follow this tutorial, you need a computer. It should have a web browser and an internet connection.

	You should know how to use a computer. Knowing how software works will also help you.
futuristic cityscape with robots and humans interacting, representing advanced technology and intelligence	**Artificial Intelligence**, or **AI**, is a big part of computer science. Its goal is to make smart machines. These machines can do tasks that usually need human thinking. Scientists have been thinking about this since the 1950s.
a vast, sparkling universe with a central glowing sphere representing AI, surrounded by smaller planets and galaxies symbolizing different AI applications like language, vision, and problem-solving	**AI** is like a big universe for making computers smart. It includes things like understanding language, seeing images, and solving hard problems. **AI** tries to make machines think and act like humans.
a stylized brain icon connected to gears and data streams, representing machine learning as a subset of AI	Now, let us talk about **Machine Learning**, or **ML**. **Machine Learning** is a part of **AI**. It is a strong way to make computers intelligent.
a split image showing two distinct approaches: on one side, a human writing explicit code with many rules; on the other side, a computer processing a large dataset to discover patterns	In old programming, we give the computer many rules. In **ML**, we give it a lot of data. The computer learns the rules from this data by itself. It finds patterns from examples.
a cartoon cat with clearly defined, sharp features, next to a flowchart showing rigid 'if-then' rules for identification, emphasizing the fragility of the rules	Let us see an example. Imagine you want to tell a computer what a cat is. In old programming, you write rules. For example: 'IF it has pointy ears, AND whiskers, THEN it is a cat.' But what if the cat's ears are folded? This rule will not work.
a diverse collage of many different cat pictures (various breeds, poses, lighting) feeding into a stylized computer brain, showing it learning autonomously	With **Machine Learning**, it is different. You show the computer thousands of cat pictures. The **ML** program sees all these pictures. It learns what a cat looks like by itself. This way is more flexible and correct.
a futuristic, glowing neural network diagram with many interconnected nodes, symbolizing advanced learning	Now, let us learn about **Deep Learning**. **Deep Learning** is a very advanced type of **Machine Learning**. It has made **AI** much better.
an abstract illustration combining a human brain silhouette with a complex, glowing neural network overlay, highlighting the inspiration	**Deep Learning** uses something called a **neural network**. This network works like the human brain. Our brain has many layers of cells called **neurons**. **Deep Learning** networks copy this idea.
a diagram showing multiple distinct layers stacked vertically, each layer filled with interconnected 'neurons' or nodes, illustrating the 'deep' aspect of the network	These **neural networks** have many layers of '**neurons**' one on top of another. Because of these many layers, we call it 'deep'. This helps it learn very complex patterns from data. It learns better than normal **ML**.

a collage showing diverse applications: a self-driving car on a road, a person interacting with a voice assistant, and abstract generative AI art	**Deep Learning** helps many new technologies we use daily. It makes self-driving cars see people. It helps voice assistants understand what you say. It also powers **generative AI**. **Deep Learning** changes many things.
	In this tutorial, we learned: **AI** makes computers smart. **Machine Learning** is a part of **AI** that learns from data, not from fixed rules. **Deep Learning** is an advanced **ML**. It uses many layers like a brain. It helps many modern **AI** tools.
	For your homework, find one real-world example of **AI**, **ML**, or **Deep Learning**. Choose an example not given in this tutorial. Think about how it helps us daily. Also, think why it is **AI**, **ML**, or **Deep Learning**.
	This **Spoken Tutorial** is from **EduPyramids Educational Services Private Limited**, **SINE**, **IIT Bombay**. Thank you for watching!